

ANNISTON, AL, USA

WMO: 722287

Lat: 33.587N

Lon: 85.856W

Elev: 594

StdP: 14.38

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 13871

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	20.9	24.8	6.5	7.9	28.8	10.8	9.7	33.0	19.6	49.6	17.9	49.1	4.4	350	0.377

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.0	94.9	74.7	92.7	74.6	90.7	74.4	78.4	88.6	77.5	87.4	76.6	86.1	6.5	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.5	136.5	82.6	74.8	133.1	81.8	73.9	129.0	80.7	42.2	88.7	41.2	87.5	40.3	86.2	84.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.8	14.4	12.4	14.2	97.5	4.2	2.8	11.2	99.6	8.7	101.2	6.3	102.8	3.3	104.8
			12.5	80.3	4.1	1.5	9.6	81.4	7.2	82.3	4.9	83.2	2.0	84.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	63.2	44.2	48.2	55.3	62.8	70.6	77.3	80.1	79.8	74.6	63.9	53.3	47.0
	DBStd	14.72	10.26	9.48	8.88	7.02	5.89	3.79	2.98	3.49	5.51	7.88	8.36	9.36
	HDD50	670	242	136	51	4	0	0	0	0	0	4	59	174
	HDD65	2623	647	472	317	121	23	0	0	0	6	117	360	560
	CDD50	5479	61	86	214	388	640	820	933	924	738	436	158	81
	CDD65	1957	1	3	15	56	197	370	468	459	294	83	9	2
	CDH74	17703	2	14	154	568	1749	3323	4389	4311	2469	668	51	5
	CDH80	7308	0	0	18	102	575	1434	2014	2005	1001	154	5	0
(14) Wind	WSAvg	5.1	6.0	6.2	6.2	5.7	4.8	4.3	4.1	3.9	4.5	4.7	5.0	5.7
(15) Precipitation	PrecAvg	52.2	5.0	4.7	6.1	5.0	4.5	4.3	4.4	3.7	3.5	2.7	4.0	4.4
	PrecMax	69.9	12.7	9.0	18.1	17.3	11.5	9.3	12.2	12.7	7.9	6.3	10.3	10.3
	PrecMin	27.9	0.9	1.2	1.4	0.2	1.8	0.0	0.8	0.5	0.3	0.0	0.9	0.7
	PrecStd	9.2	2.2	2.2	3.7	3.1	2.2	2.4	2.1	2.4	2.1	1.7	2.0	2.1
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	72.9	76.3	82.4	86.0	91.1	95.4	97.3	99.0	95.5	87.4	79.5	74.2
		MCWB	61.8	63.4	65.0	67.8	72.4	73.3	75.5	76.2	71.5	68.5	64.7	65.1
	2%	DB	69.1	71.8	78.3	82.9	88.2	92.6	94.1	95.0	92.1	83.9	74.8	70.4
		MCWB	60.7	61.3	62.7	65.9	71.3	73.9	75.8	75.7	71.7	68.5	62.9	62.9
	5%	DB	65.2	68.4	74.8	80.4	86.0	90.4	91.8	92.3	89.2	81.1	71.8	66.8
		MCWB	58.9	59.2	60.8	64.8	70.4	74.0	75.8	75.1	71.9	66.4	61.9	60.5
	10%	DB	61.4	64.5	71.0	77.4	83.4	88.1	89.8	89.9	86.1	77.8	68.6	62.9
		MCWB	55.6	56.2	59.1	63.4	69.4	73.5	75.5	74.7	71.6	65.3	60.2	57.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	65.6	66.4	68.2	71.8	76.0	78.4	80.2	80.1	76.5	74.4	70.3	67.8
		MCDB	70.6	73.1	75.3	81.1	85.6	88.2	90.1	92.1	85.6	81.9	73.6	72.2
	2%	WB	62.8	64.2	65.8	69.4	74.2	77.2	78.5	78.4	75.3	72.2	67.2	64.8
		MCDB	66.7	69.8	74.0	77.6	83.6	87.3	88.3	89.0	84.3	79.2	71.5	68.6
	5%	WB	60.0	61.9	63.9	67.8	72.7	76.2	77.6	77.4	74.4	70.5	64.6	61.9
		MCDB	64.2	66.7	71.4	75.9	81.8	86.0	87.5	87.3	82.9	76.7	69.9	65.8
	10%	WB	56.7	58.5	61.5	66.0	71.3	75.2	76.8	76.4	73.3	68.6	61.5	58.4
		MCDB	60.9	62.9	68.6	73.6	79.9	84.5	86.2	85.6	81.2	74.3	66.8	62.3
(35) Mean Daily Temperature Range	5% DB	MDBR	20.3	21.1	22.5	23.6	21.5	19.9	19.0	19.3	20.5	22.6	23.0	19.8
		MCDBR	23.6	24.5	26.8	26.8	24.4	23.7	22.1	23.3	24.5	25.0	25.3	22.3
		MCWBR	16.2	15.5	13.7	12.2	10.0	8.7	7.6	7.4	8.6	11.2	14.5	15.7
	5% WB	MCDBR	19.5	20.1	21.2	20.6	20.3	19.7	19.1	19.3	18.3	17.8	19.0	19.1
		MCWBR	15.3	15.4	12.9	11.4	9.3	8.5	7.6	7.1	7.6	9.6	13.6	15.7
(40) Clear-Sky Solar Irradiance	taub		0.321	0.325	0.362	0.394	0.469	0.488	0.537	0.538	0.433	0.374	0.344	0.324
	taud		2.508	2.494	2.413	2.307	2.136	2.131	2.008	1.993	2.278	2.430	2.453	2.515
	Ebn at Noon		284	294	290	284	262	256	243	240	263	273	272	276
	Edh at Noon		27	30	35	41	49	49	56	56	40	32	28	25
(44) All-Sky Solar Radiation	RadAvg		838	1004	1361	1745	1905	1942	1873	1730	1510	1271	942	712
	RadStd		57	130	122	117	159	152	129	99	141	166	101	74

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+182	+126
Regional (47 neighbors)	+0.88	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+286	+203

Nomenclature: See separate page